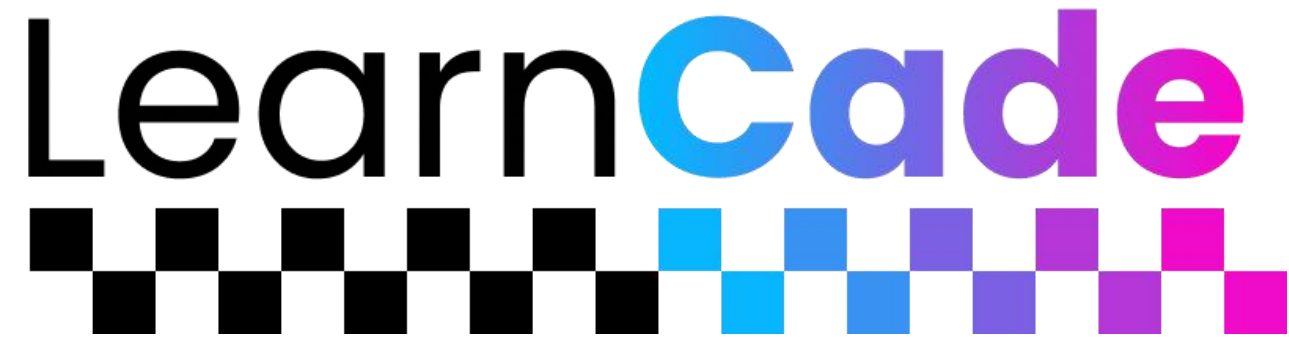




Case Study: LearnCade

“A platform for generating classic arcade games into engaging, educational experiences for kids.”



Additionally, LearnCode focuses strongly on the User Experience (UX) and the User Interface (UI) to ensure a smooth, engaging, and accessible flow for all users.



- **Balancing Education and Entertainment:** The core idea of LearnCade is to combine learning with fun through play-based learning. However, finding the right balance between the two can prove challenging.
- **Platform Design:** Designing a platform that appeals to both children and teachers can be tricky as if the platform feels too childish, adults may find it frustrating to navigate, but it feels too mature, kids may lose interest.



- Utilise LLMs to innovatively assist the creation of educational games
- Become a one-stop educational platform that kids are excited to use.
- To be used by teachers in a classroom environment providing them an alternative way of teaching their students.

2. User Persona's

“I learn best when it is enjoyable, not boring”

User Persona 1: Leo

Background:

- Age: 7 years old
- Education: Year 2 (Primary School)
- Family: Lives with Mum, Dad and his older brother
- Tech Access: Uses Mum's tablet or family laptop.

Goals and motivations

- Discover new and fun ways to learn
- Expand his knowledge and skills through fun, interactive activities
- Create video games quickly and easily

Key characteristics

- Passionate about exceling in school and getting good grade to impress his parents
- Enjoys arcade games as he grew up playing them with his dad
- Always curious and eager to learn new things

Behaviours and preferences

- Leo spends two hours playing games each day on his mum's tablet and/or family laptop
- He prefers short, exciting challenges over long sessions
- He is highly creative and enjoys games with strong customisation options
- Leo enjoys participating in classroom activities.

Needs and pain points

- Needs instant feedback as he wants to know if he's on the right track.
- Wants a sense of achievement as he likes rewards
- Seeking game-like learning because traditional methods are boring to him

2. User Persona's

“I want my students to love learning, make them curious and excited”

User Persona 2: Ms Grace

Background:

- Age: 30 years old
- Occupation: Year 2 Teacher (Primary School)
- Subjects: English, Spelling, Literacy
- School: Primary school with mixed-ability classes
- Tech Access: Classroom smartboard, student tablets and personal laptop

Goals and motivations

- Enhance her teaching by finding different ways to teach her class.
- Encourage her students to enjoy learning
- Connect with her students and make them feel exciting to attend school

Behaviours and preferences

- Ms Grace prefers using new technology over traditional methods
- She often researches fun and unique ways to teach her students in her free time.
- She likes to share great learning tools with other teachers during her breaks.
- She appreciates when an application is user-friendly and straightforward.

Key characteristics

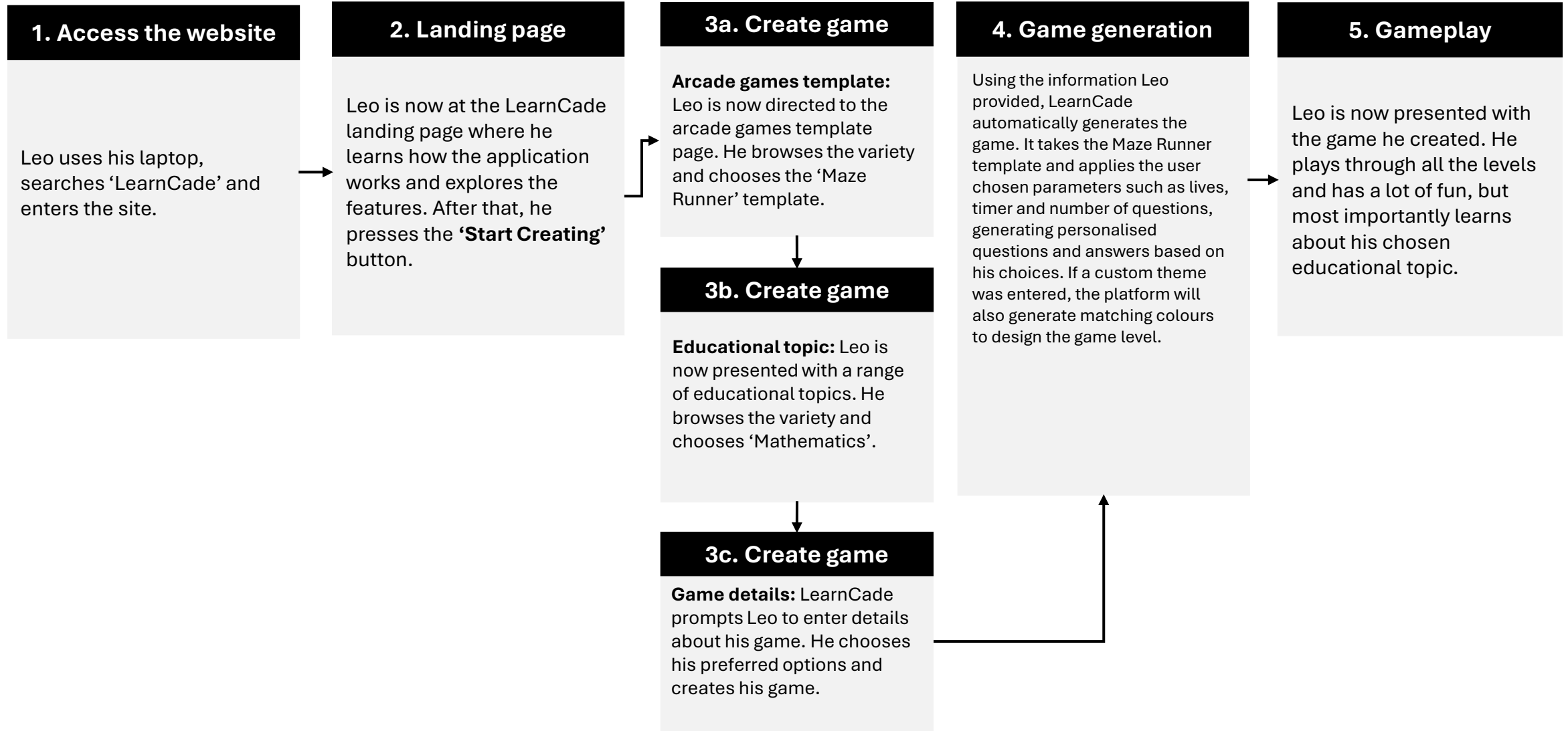
- Passionate about creating memorable experiences for her students
- Eager to start adapting new technologies into her classroom
- Seeks a balance between education and fun

Needs and pain points

- Needs a range of relevant educational topics to choose from
- Seeks an easy way to create games without requiring game development tools and skills
- Wants to use technology that can be easily setup for her students
- Needs activities that work for different ability levels.

3. User Flow

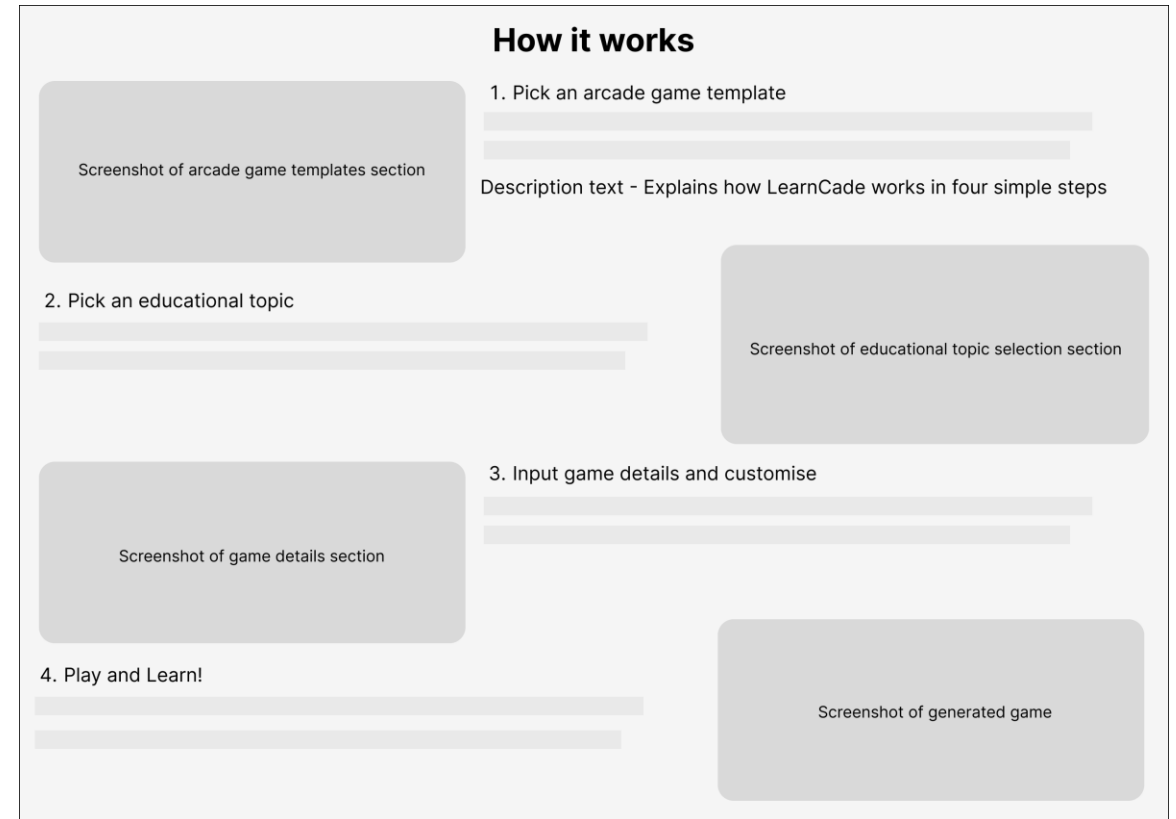
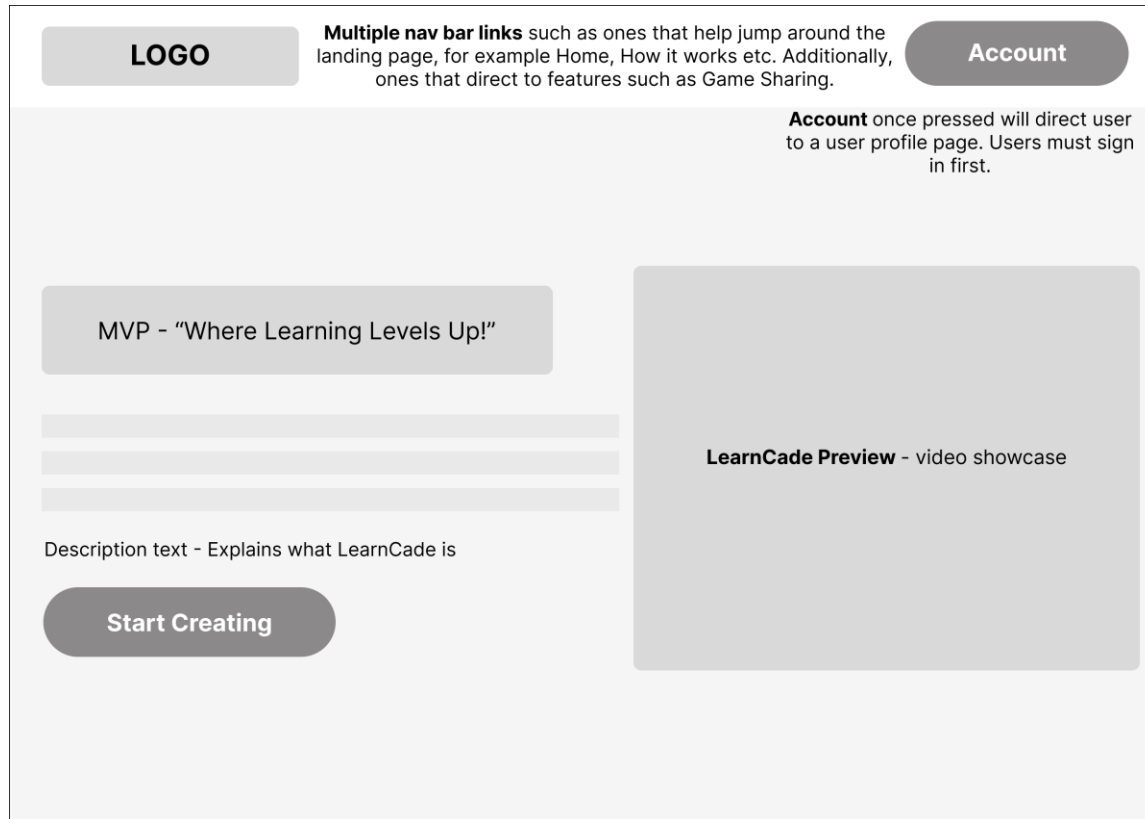
The following user flow provides a step-by-step overview of **Leo's** interactions as he engages with the LearnCade website:



4. Wireframes

The following low-fidelity (low-fi) wireframes have been created to visualise the user interface for LearnCade. The designs have been made to fulfil the needs of the Project Requirements, User Persona's and User Flow.

Landing page:

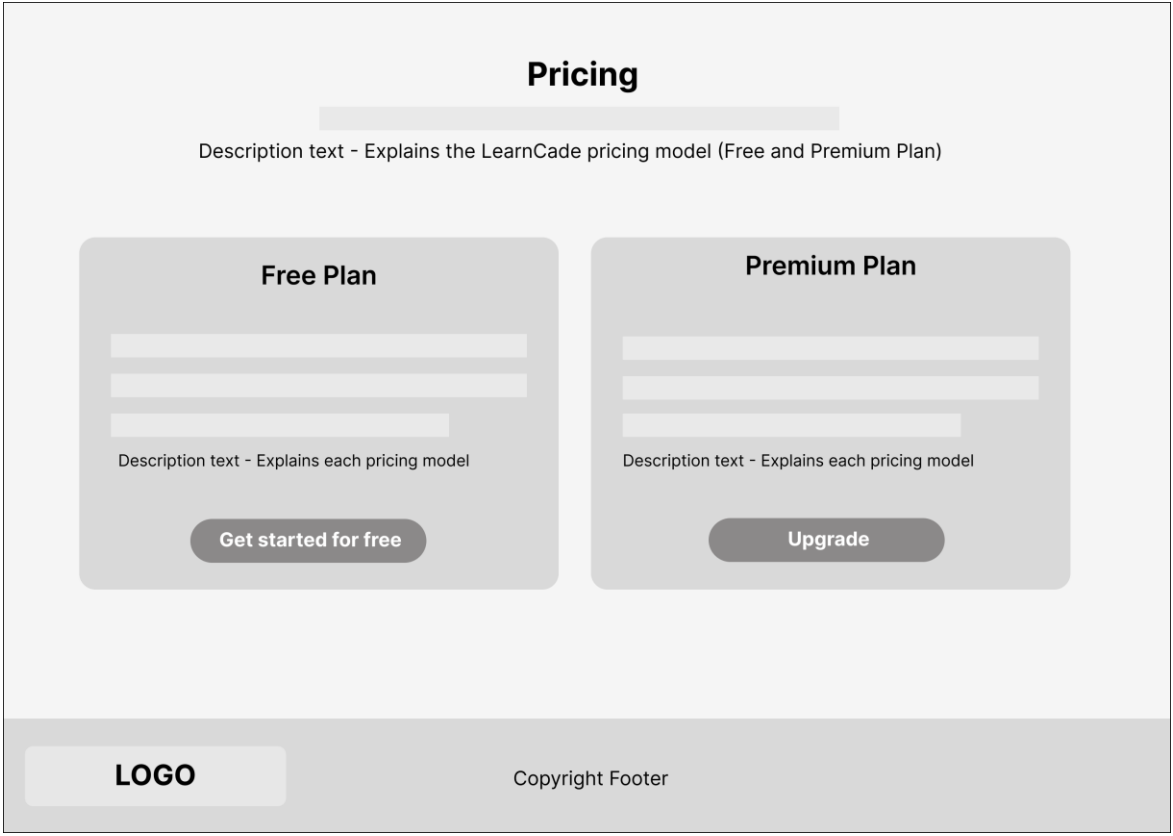
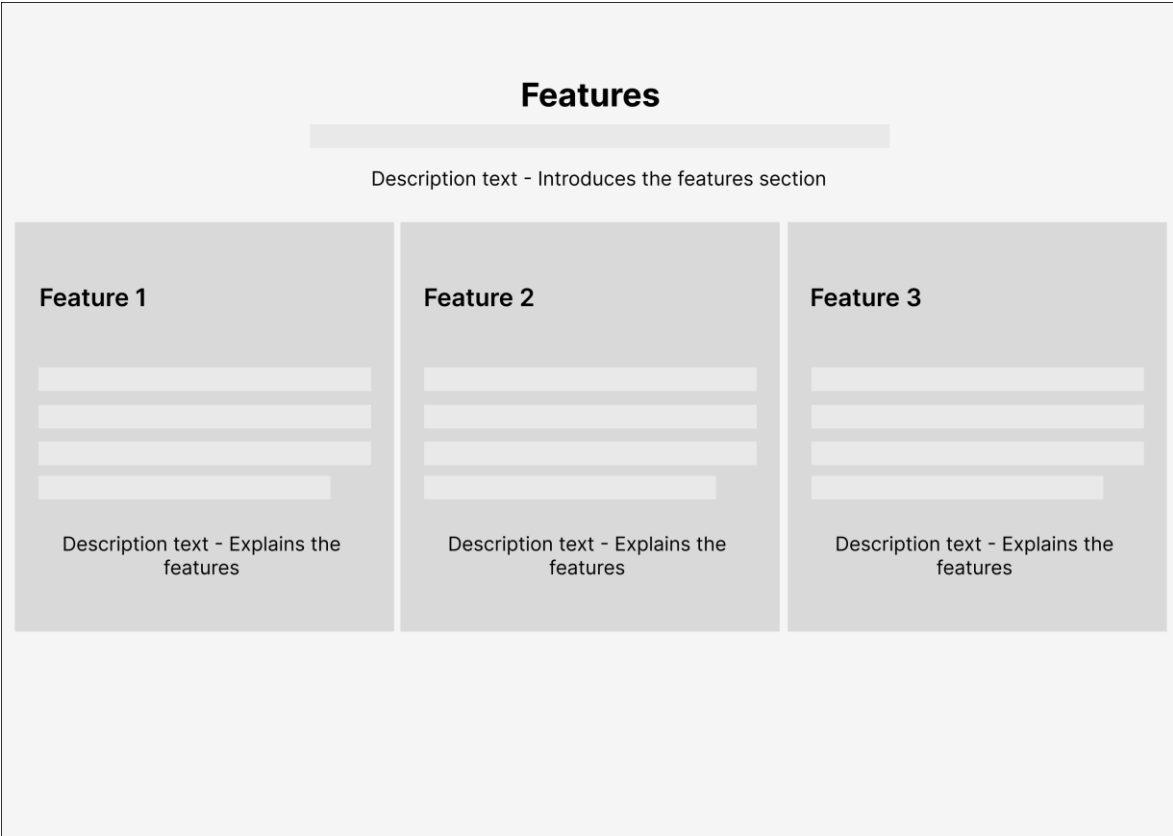


Wireframes created using Figma

4. Wireframes

The following low-fidelity (low-fi) wireframes have been created to visualise the user interface for LearnCade. The designs have been made to fulfil the needs of the Project Requirements, User Persona’s and User Flow.

Landing page (continued):

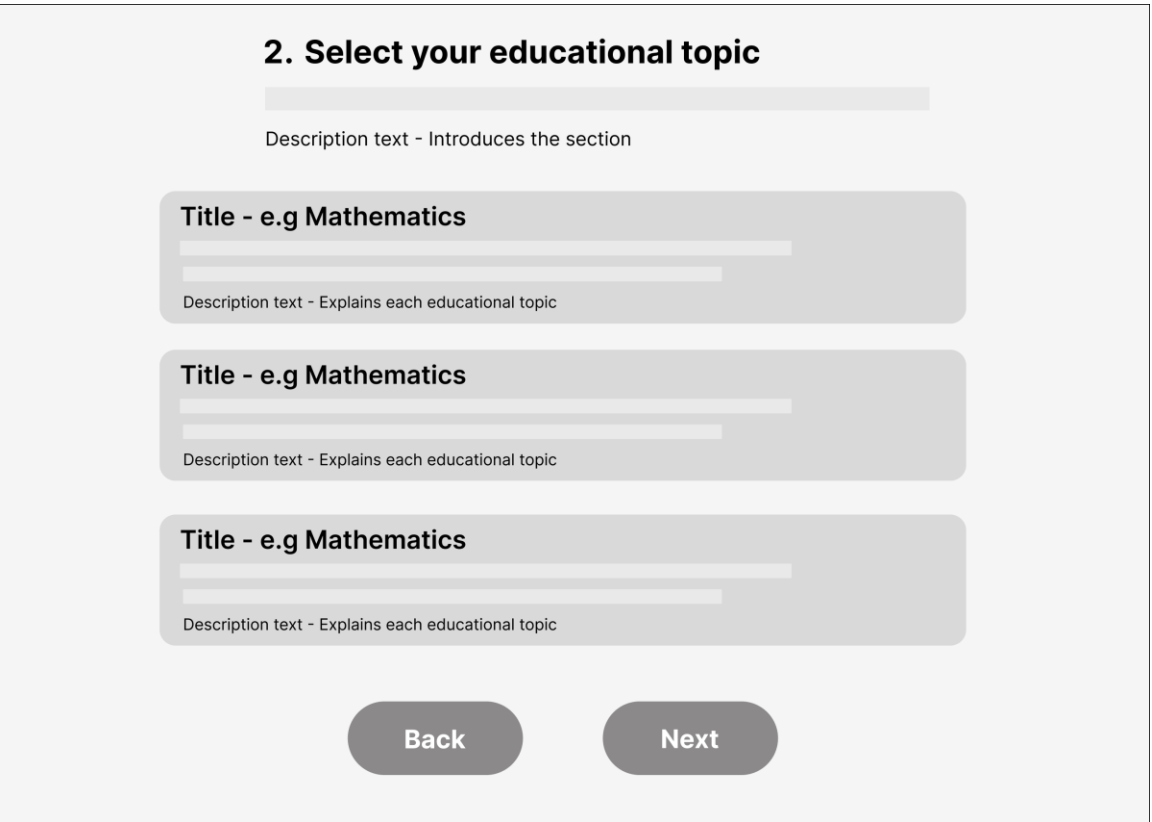
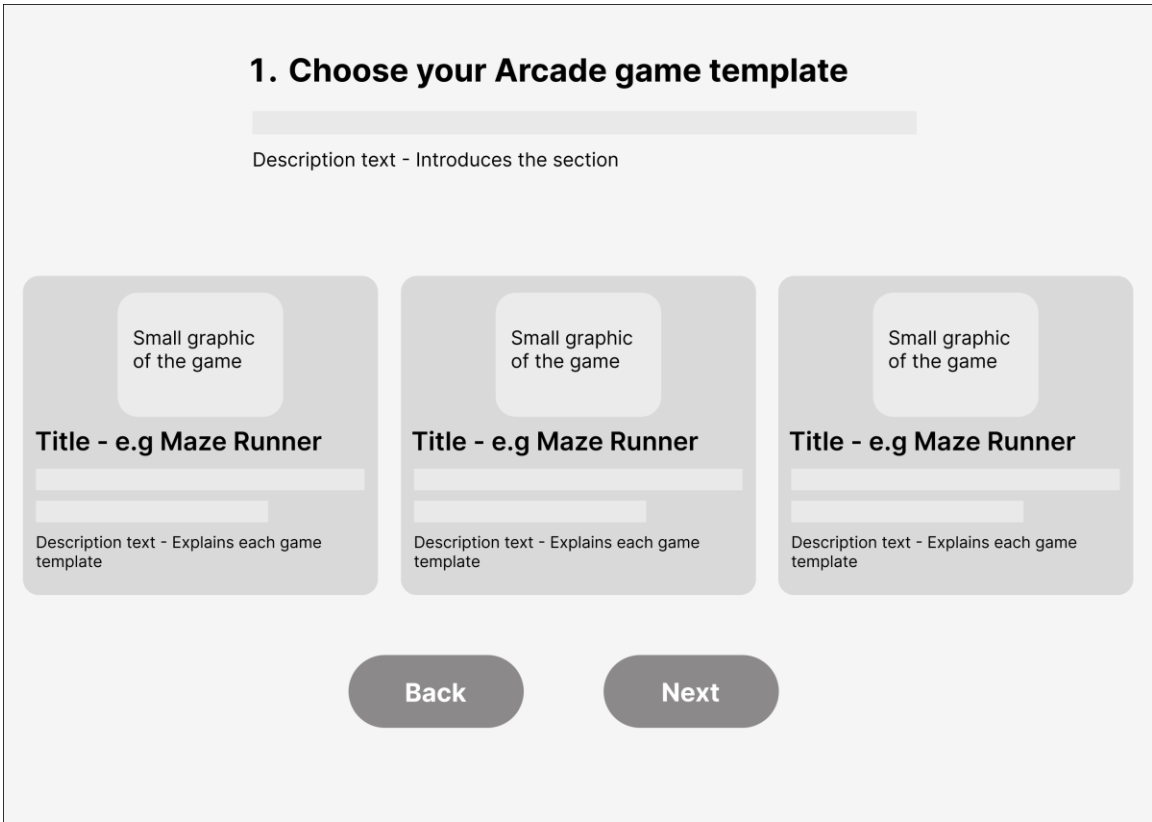


Wireframes created using Figma

4. Wireframes

The following low-fidelity (low-fi) wireframes have been created to visualise the user interface for LearnCade. The designs have been made to fulfil the needs of the Project Requirements, User Persona’s and User Flow.

Create pages:



Wireframes created using Figma

4. Wireframes

The following low-fidelity (low-fi) wireframes have been created to visualise the user interface for LearnCade. The designs have been made to fulfil the needs of the Project Requirements, User Persona's and User Flow.

Create pages (continued):

3. Enter details about your game

Description text - Introduces the section

Option 1

Dropdown box - Users will be able to choose options to customise their game. For example, if the chosen topic was Mathematics, Option 1 may state 'Choose Operation'. The dropdown box will include Addition, Subtraction, Multiplication and Division.

Option 2

Custom option 1

Text box - Users will be able to type in here
Custom option 1 may state 'custom theme'

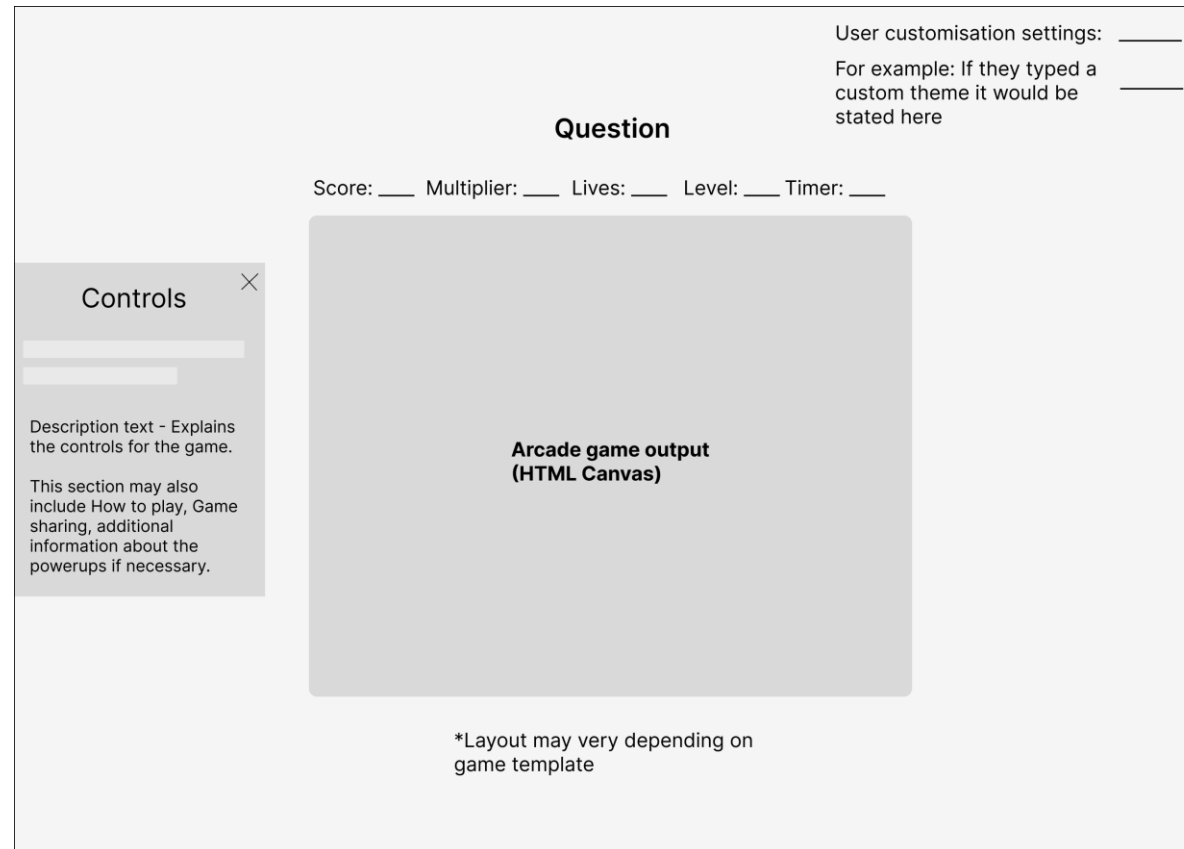
Create game

Wireframes created using Figma

4. Wireframes

The following low-fidelity (low-fi) wireframes have been created to visualise the user interface for LearnCade. The designs have been made to fulfil the needs of the Project Requirements, User Persona's and User Flow.

Play page:

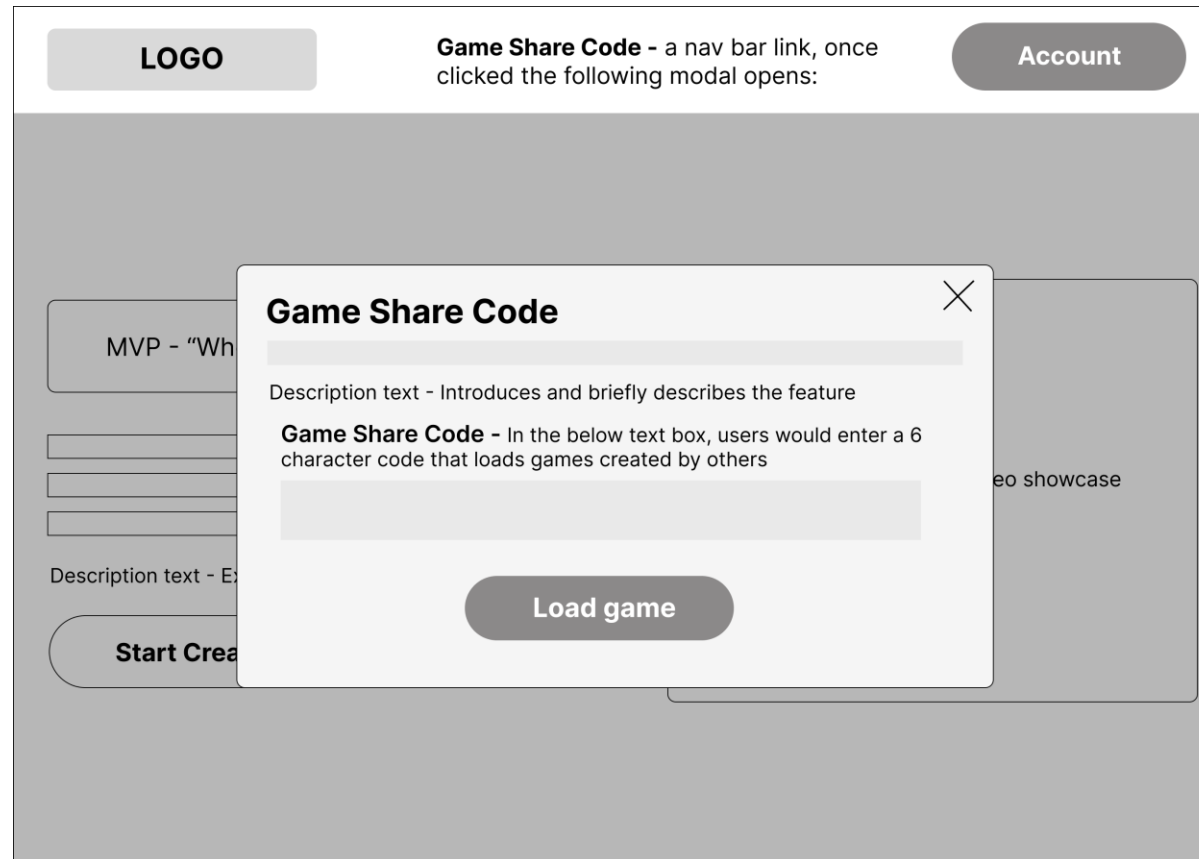


Wireframes created using Figma

4. Wireframes (Additional features)

The following low-fidelity (low-fi) wireframes are made for the additional features within LearnCade. These features will enable Game Sharing, Progress Tracking and a local multiplayer.

Game Share Code:



Wireframes created using Figma

4. Wireframes (Additional features)

The following low-fidelity (low-fi) wireframes are made for the additional features within LearnCade. These features will enable Game Sharing, Progress Tracking and a local multiplayer.

Account Profile:

Username
Additional information, for example the email used to make the account, when the account was created etc.

Overall Score: Games Played: High Score:

Score earned - In this section of the profile page, the user can filter the scored they have in the last 24 hours, 7 days, 30 days and all time.

Best game - This section will display the game where the user got their high score

Recent games - This section of the profile will show the last 10 games created/played by the user. They can replay any of the games saved in the profile.

Username
Additional information, for example the email used to make the account, when the account was created etc.

Overall Score: Games Played: High Score:

Score earned - In this section of the profile page, the user can filter the scored they have in the last 24 hours, 7 days, 30 days and all time.

Best game - This section will display the game where the user got their high score

Recent games - This section of the profile will show the last 10 games created/played by the user. They can replay any of the games saved in the profile.

Game title - e.g Maze Runner ✕
Game info - e.g. Mathematics - Multiplication - 5x times table
Score - e.g. 5,000

Timer: Lives:

Questions: Custom theme:

Game Share Code - A game share code will be displayed here, it will be a 6 digit code. They can share this code with others.

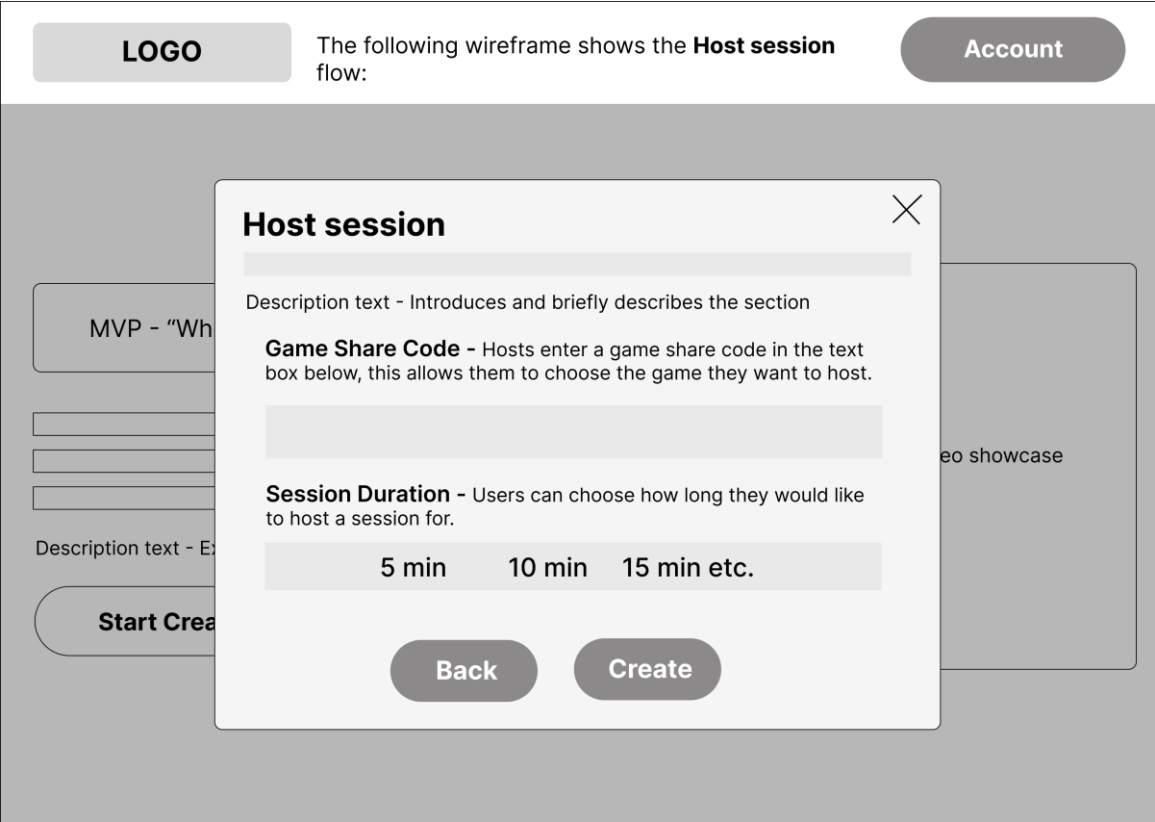
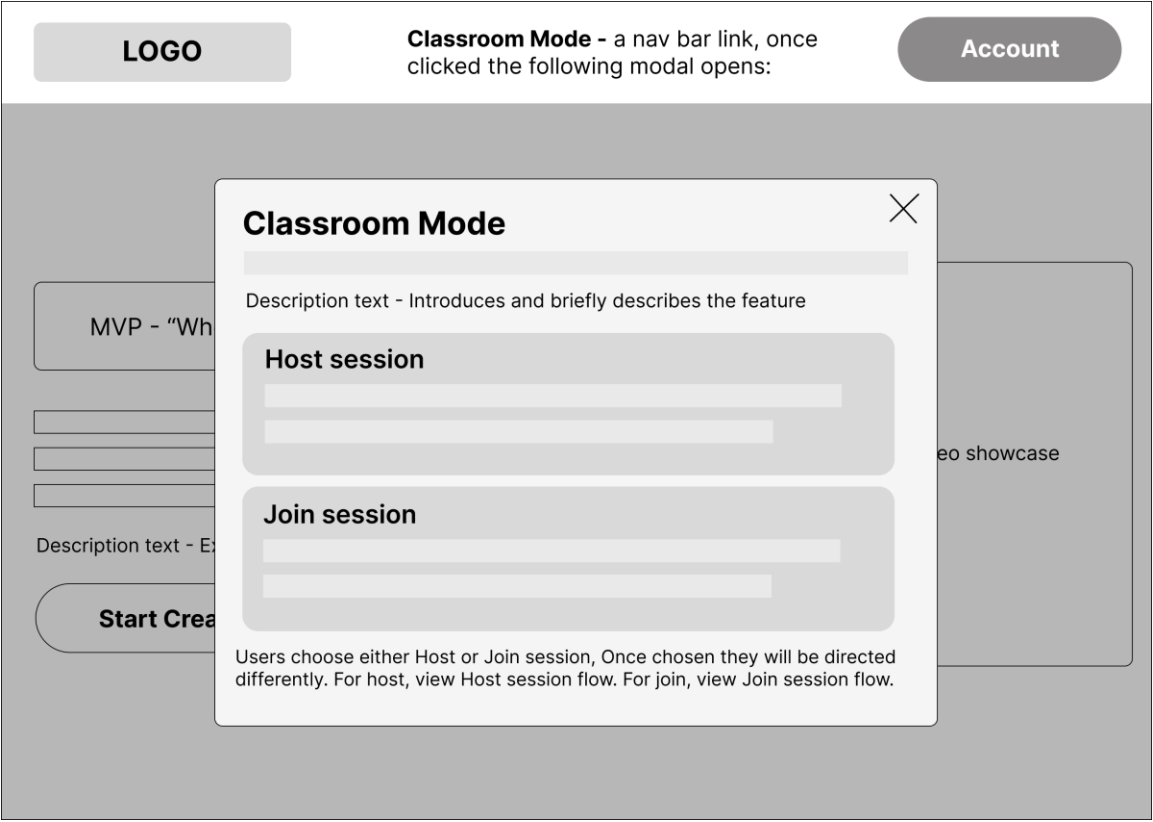
Play again

Wireframes created using Figma

4. Wireframes (Additional features)

The following low-fidelity (low-fi) wireframes are made for the additional features within LearnCade. These features will enable Game Sharing, Progress Tracking and a local multiplayer.

Classroom mode:

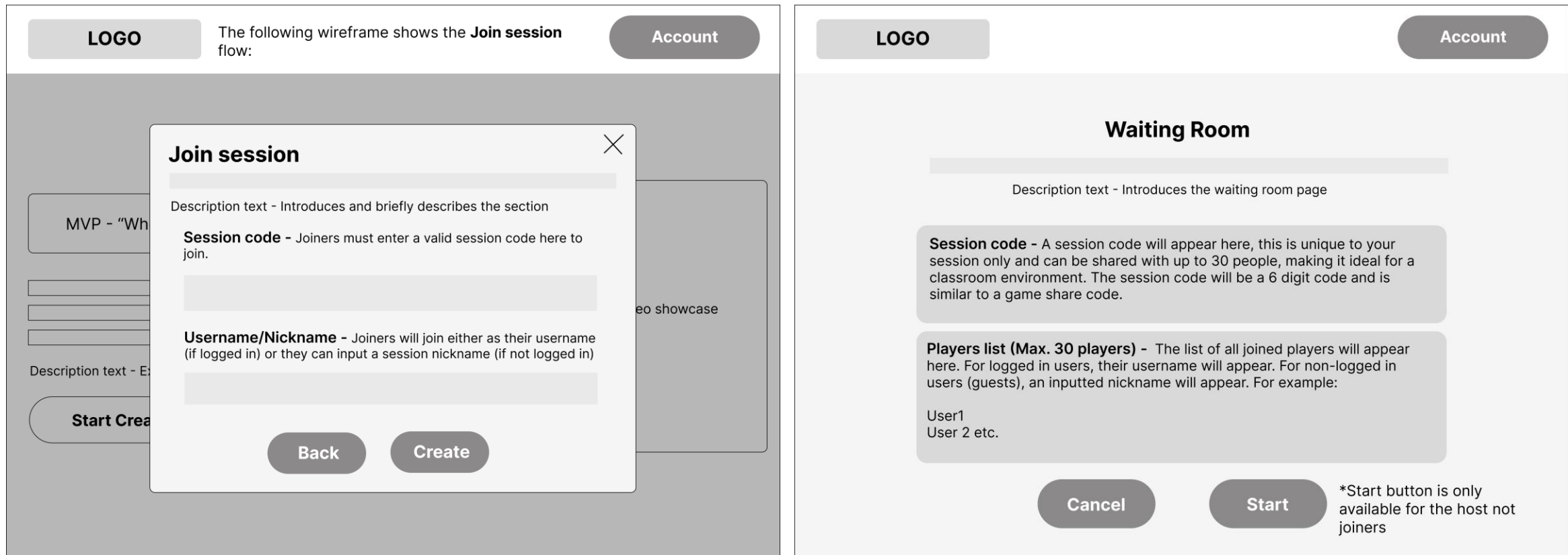


Wireframes created using Figma

4. Wireframes (Additional features)

The following low-fidelity (low-fi) wireframes are made for the additional features within LearnCade. These features will enable Game Sharing, Progress Tracking and a local multiplayer.

Classroom mode (continued):

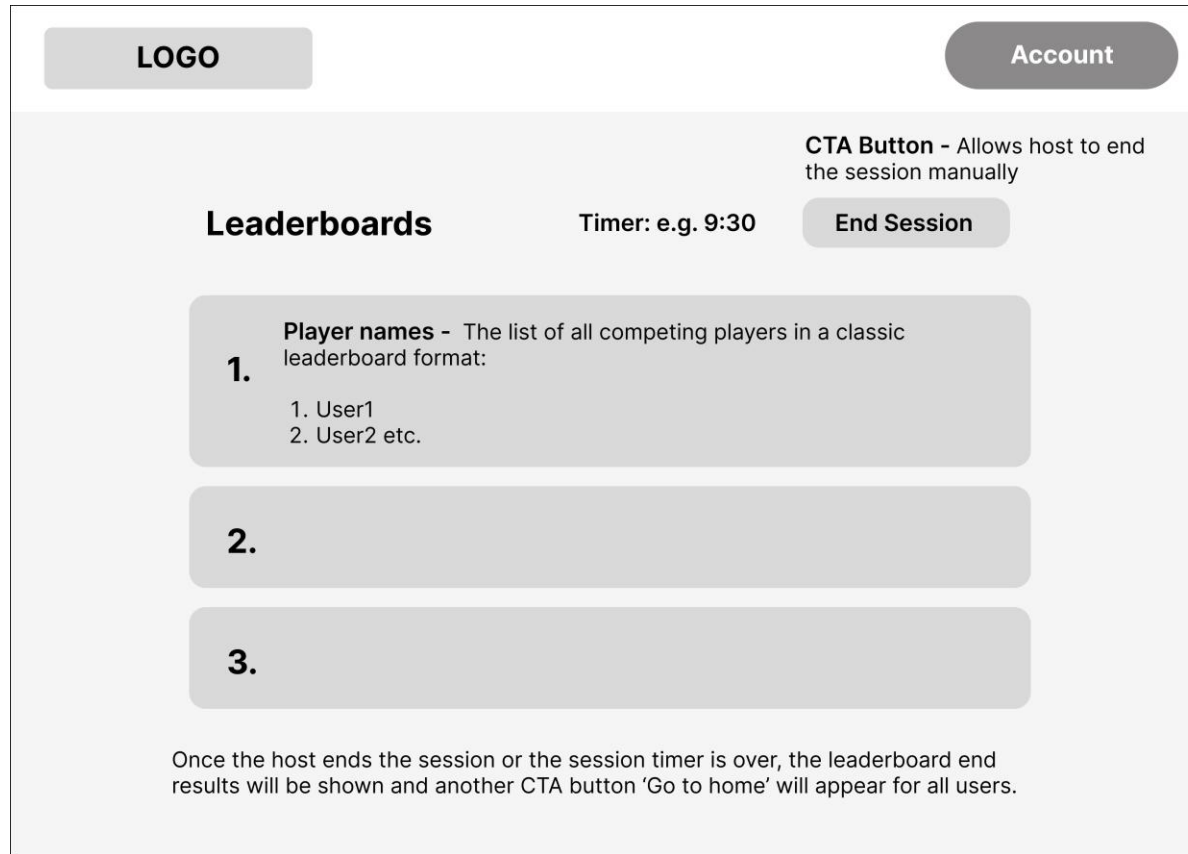


Wireframes created using Figma

4. Wireframes (Additional features)

The following low-fidelity (low-fi) wireframes are made for the additional features within LearnCade. These features will enable Game Sharing, Progress Tracking and a local multiplayer.

Classroom mode (continued):



Wireframes created using Figma

6. Game Design

Arcade Game 1: Maze Runner

For the first game, I looked to the mechanics of the classic arcade game, Pac-Man. It's core loop of navigation, evasion and collection provides an effective framework for testing knowledge under pressure. Additionally, I believe that this type of game is essential for LearnCade as it will give users that strong arcade feeling.



Figure 1 (Pac-Man)

Image source - <https://thereader.mitpress.mit.edu/how-pac-man-revolutionized-gaming/>

The Adaptation

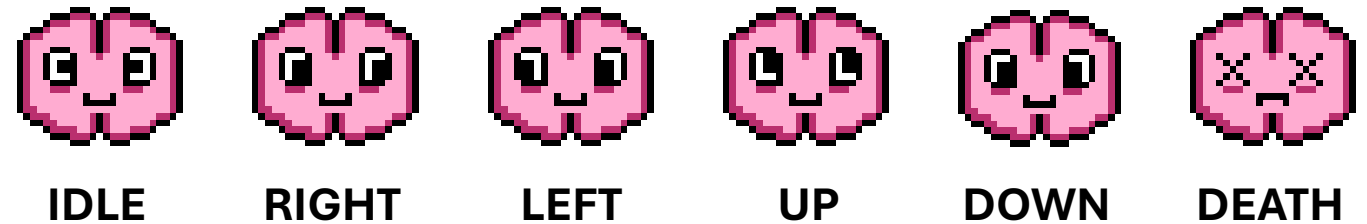
Figure 1 shows the original Pac-Man game, which involves the user avoiding ghosts and eating dots to earn score and eventually beat the level by eating them all. While inspired by the original, Maze Runner pivots the mechanics to ensure a unique experience and a strong focus on education:

1. Instead of multiple small dots that give score, Maze Runner will have 4 circles which contain answers to a given question. Three of these will be incorrect, one will be correct. The user must use their knowledge to beat the level and earn score.
2. The Pac-Man character will be adapted for LearnCade. Instead of the iconic yellow eater, we will use a friendly brain-shaped character, which heavily links to the education aspect.
3. Maze Runner will also have a multiplier system to encourage users to answer correctly. For example, Level 1 – 1.0x, if they answer correctly, Level 2 – 1.1x, Level 3 – 1.2x etc. However, if they answer incorrectly the multiplier will be reset, encouraging users to master topics to earn the best score possible.
4. Maze Runner will maintain the traditional movement mechanics (WASD or keyboard arrows)
5. Maze runner will have a reduced number of ghosts with less aggression to ensure that the users' focus is on the question. It will include a chase mode if users get too close to keep it tense and challenging.
6. Maze Runner will also include powerups around the maze to allow users to better their score but also encourage them to explore the maze.

6. Game Design

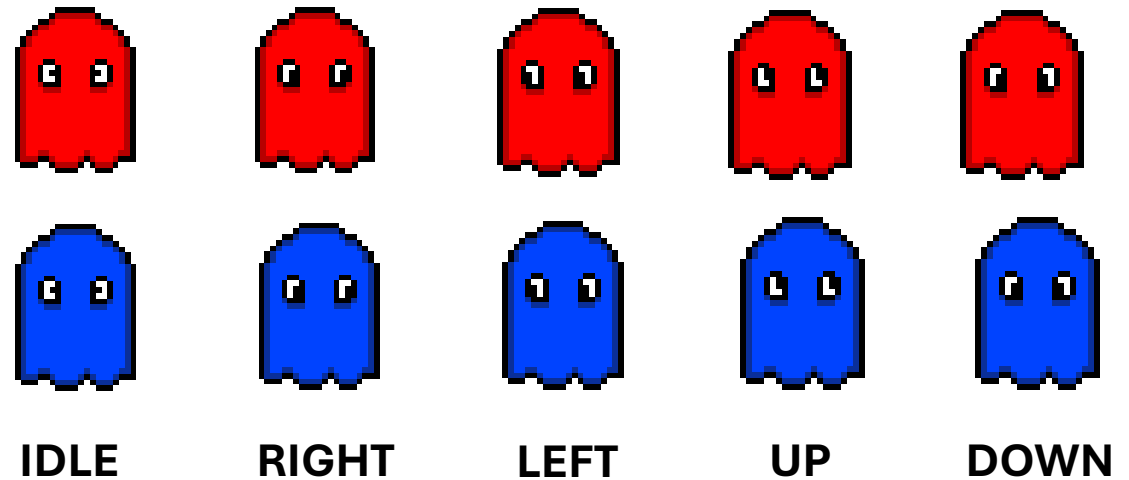
Maze Runner: Brain character (User):

The following 32x32 sprite images showcase the playable character in LearnCade's Maze Runner. As you can see, the character design focuses on appealing to the primary target audience by being fun, vibrant and playful but also links to the aspect of education. Additionally, I have created six different possible frames showcasing idle, variety of movement and death.



Maze Runner: Ghost character (NPC)

The following 32x32 sprite images present the non-playable characters in LearnCade's Maze Runner. This character is designed after the traditional ghost-like enemies in the original Pac-Man game and will be used to make the game feel tense and fast-paced. They will also have different colours to make the game more colourful and appealing to our target audience.



Assets created using Krita

6. Game Design

Maze Runner:

The following screen captures the Maze Runner arcade game within LearnCade. It involves controlling a brain-like character to collect an answer to a given question, while avoiding detection from patrolling ghosts. In this case, the correct answer is 15, therefore, if the user collects that answer, they will beat the level and earn score. Additionally, there are two +100 powerups that allow users to increase their score by 200 per level. However, it is risky as it costs time and increases the chance of being spotted by the ghosts.

If users answer incorrectly by collecting the incorrect answer, they will lose a life and will be able to retry the level. However, if they have no lives left the game will end. Another way they can lose a life is by being caught by the ghosts. Finally, the last way to die is by running out of time.

Overall, the product serves as a fun and highly educational game for our target audience.

In the example shown, the user has chosen Maze Runner, Mathematics, Multiplication, 5x times table, 3 lives, 10 questions/levels, a 40 second timer and the default theme.



6. Game Design

Arcade Game 2: Space Shooter

For the second game, I looked to the mechanics of Space Invaders. This game involves shooting, dodging, and eliminating incoming threats which creates an action-focused framework that rewards accuracy, reflexes, and quick decision-making. This makes it effective for LearnCade as it tests the users' ability to actively respond to questions while managing danger, combining learning with fast-paced, reactive gameplay.



Figure 2 (Space Invaders)

Image source - <https://segadoes.com/2014/07/31/space-invaders/>

The Adaptation

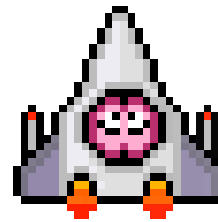
Figure 2 shows the original Space Invaders game, where players control a cannon to shoot descending aliens while avoiding incoming fire and the main goal of the game is to kill all the aliens before they reach the bottom. While inspired by the original, Space Shooter pivots the mechanics to ensure a unique experience and a strong focus on education:

1. Instead of multiple waves of aliens, Space Shooter will include only two waves of aliens that are protecting four circles which contain answers to a given question. Three of these will be incorrect, one will be correct. The user must defeat the aliens quickly and shoot the right answer and beat the level to earn score. The user cannot shoot the answer circles if all the aliens are not killed as a laser shield will stop bullets from hitting the answers. Only once the aliens are killed the laser shield will disappear.
2. The main character will be adapted for LearnCade. Space Shooter will use a moveable rocketship with a laser cannon attached. Inside will be the same brain-shaped character seen in Maze Runner. This way we can get users to form a connection to the character, making it like a mascot/avatar for the application.
3. Space Shooter will include the same multiplier system as Maze Runner.
4. Space Shooter will have the following movement mechanics: Arrows to move left and right or A/D to move left and right. Users cannot move up or down.
5. Space shooter will include the same alien-like enemies as the original, however, we will have a reduced number of aliens with less aggression to ensure that the users' focus is on the question.
6. Space Shooter will include the same powerup as Maze Runner (+100 points powerup)

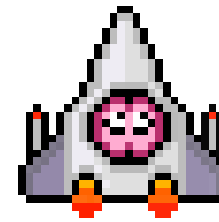
6. Game Design

Space Shooter: Rocketship character (User)

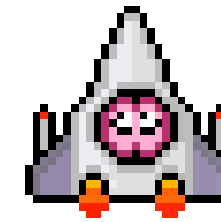
The following 32x32 sprite images showcase the playable character in LearnCade's Space Shooter. As you can identify, the character is designed to fit the theme of the game (space) and includes the brain character giving it a playful and vibrant look. I have also created four different frames showcasing idle, left, right, and death.



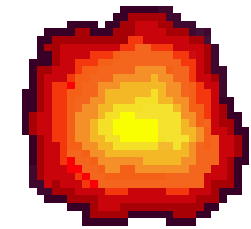
IDLE



RIGHT



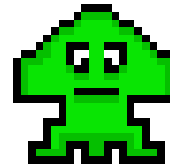
LEFT



DEATH

Space Shooter: Alien character (NPC)

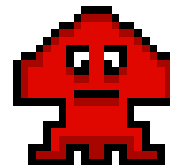
The following 32x32 sprite images present the non-playable characters in LearnCade's Space Shooter. This character is designed after the original alien enemies. I have showcased the different colours which allow the game to be more colourful.



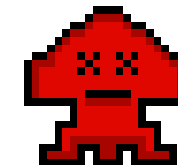
DOWN



DEATH



DOWN



DEATH

6. Game Design

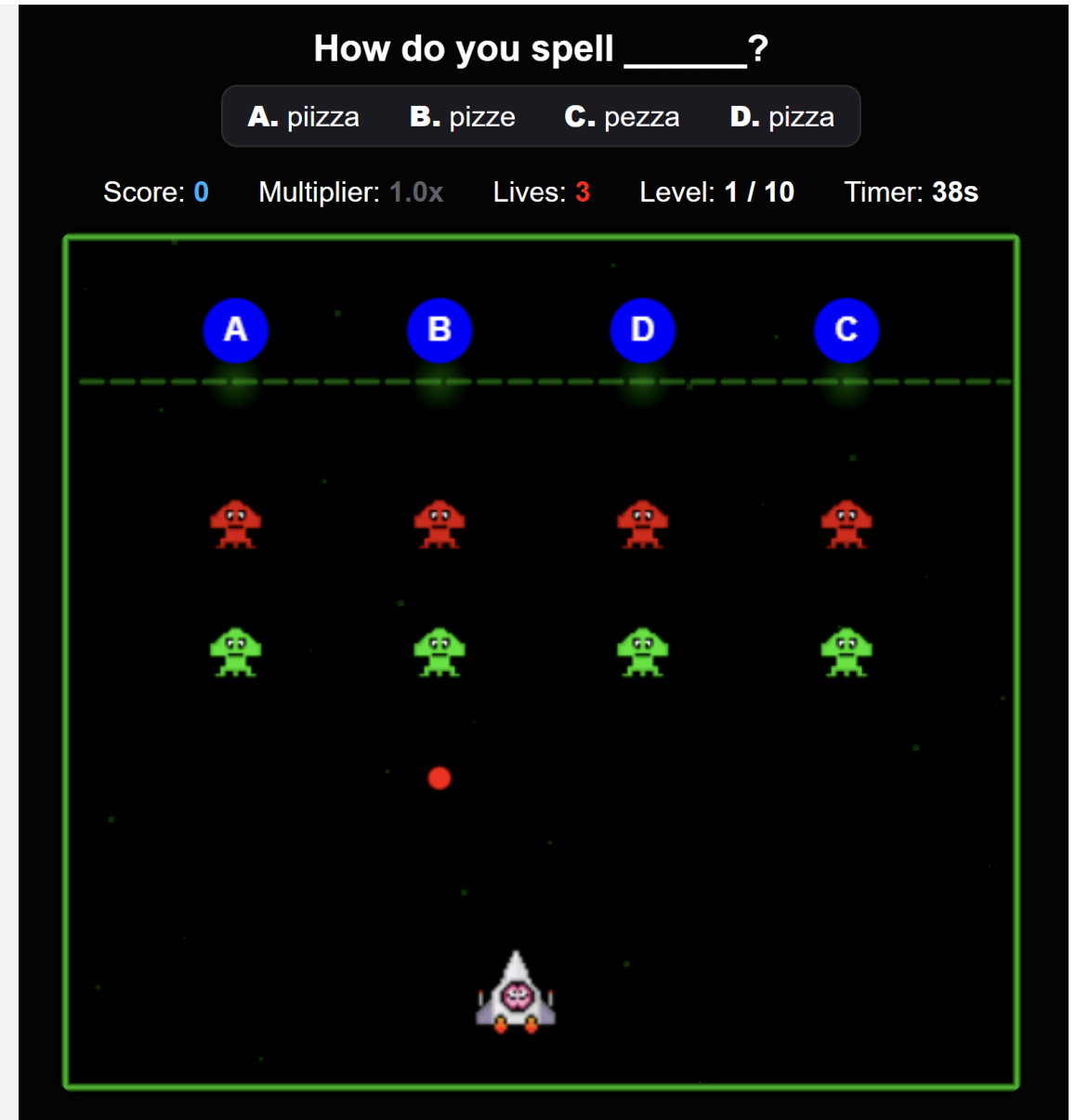
Space Shooter:

The following screen captures the Space Shooter arcade game within LearnCade. It involves controlling a rocketship to shoot aliens who are attacking you. Once the aliens are killed, the laser shielding the answer circles will disappear and the user can blast the correct answer to beat the level and earn score. In this case, the correct answer is D. pizza, if the user shoots this answer circle they will be rewarded. Each alien killed also rewards players, +10 for the green and +15 for the red aliens. Additionally, a bonus +100 powerup will spawn once the laser shield is destroyed, users can choose to shoot it to get an extra 100 points, or they can save time and focus on the answer.

If users answered incorrectly by shooting an incorrect answer circle, they lose a life and can retry the level. Additional ways of losing lives are being shot by the aliens, any of the aliens reach the bottom or just by simply running out of time, therefore, users must think quick.

Overall, it effectively combines the traditional Space Invaders concept with strong educational value and entertainment for the target audience.

In the example shown, the user has chosen Space Shooter, Spelling, Food, 3 lives, 10 levels/questions, a 40 seconds timer and a theme of 'green and blue'



6. Game Design

Arcade Game 3: Brick Breaker

For the third and final game, I looked to the mechanics of the classic arcade game Breakout. This game involves launching a ball and bouncing it with a paddle to destroy bricks, creating a strategic framework where players must carefully control the ball's trajectory. In LearnCade, the bricks act as barriers protecting the answers, encouraging players to think ahead and break through the right areas. This makes the game effective for learning as it combines problem-solving and strategy.

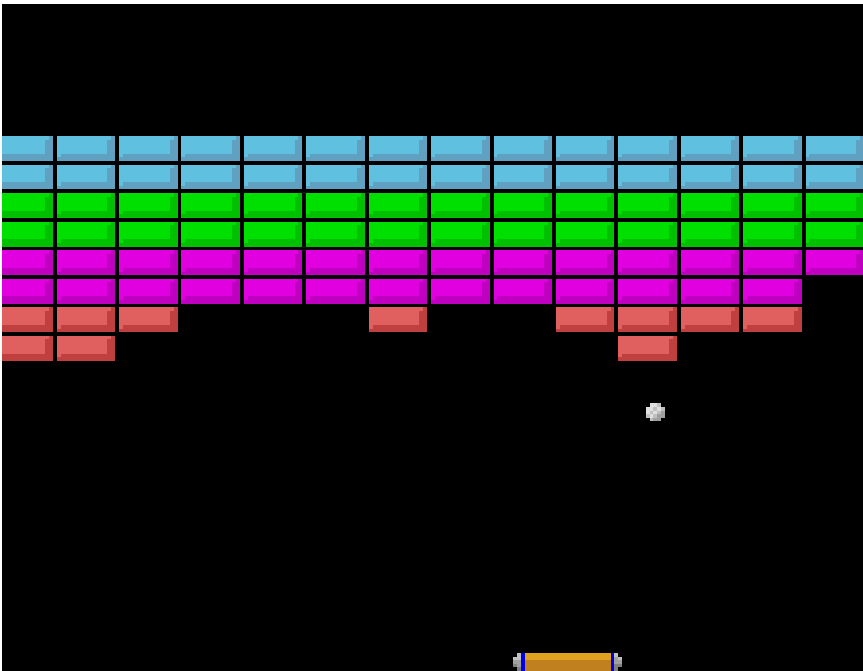


Figure 3 (Breakout)

Image source - <https://www.atarilegend.com/games/super-breakout>

The Adaptation

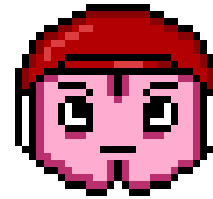
Figure 3 shows the original Breakout game, where players launch a ball and control it using a paddle to destroy a wall of bricks to earn score. The main goal is to clear the level by destroying all the bricks without dropping the ball. While inspired by the original, Brick Breaker pivots the mechanics to ensure a unique experience and a strong focus on education:

1. Brick Breaker will have four answer circles that will be placed at the top of a brick wall with multiple columns. Three of these will be incorrect, one will be correct. The user must break down the column of bricks that is holding the correct answer circle, this will make it drop and then users must act quick to collect the answer with the paddle.
2. The main character will be adapted for LearnCade. Brick Breaker will use the same brain-like character; however, it will be designed with a helmet to make it unique to this game.
3. Brick Breaker will include the same multiplier system as Maze Runner and Space Shooter.
4. Brick Breaker will include the following movement mechanics, Spacebar to launch the ball, left and right arrows or A/D to move the paddle left and right. The user cannot move up or down.
5. Unlike the original, Brick Breaker will not punish users for dropping the ball, the ball will just respawn back on the paddle, but instead the punishment will be for dropping the right answer circle or collecting a wrong answer circle.
6. Brick Breaker will feature the following powerups: +100 points powerup, and a bomb powerup which when hit, it explodes and destroys multiple bricks making it easier for users to clear the wall faster.

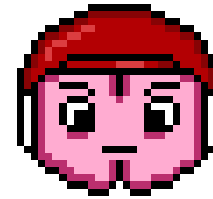
6. Game Design

Brick Breaker: Brain character with helmet (User)

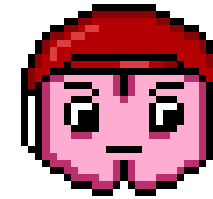
The following 32x32 sprite images showcase the playable character in LearnCade's Brick Breaker. The character is designed to fit the theme of the game as the helmet represents the character trying to protect itself from the impact of hitting the bricks. I have also created four different frames showcasing idle/up, down, right and left.



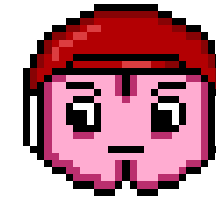
IDLE/UP



DOWN



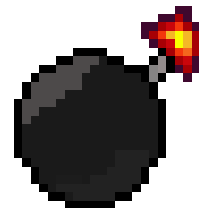
RIGHT



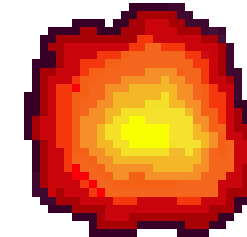
LEFT

Brick Breaker: Bomb (Powerup)

The following 32x32 sprite images present the powerup in LearnCade's Brick Breaker. As you can identify, I have designed a bomb asset which when activated, explodes and destroys multiple bricks, helping users clear the brick wall faster.



Bomb (before
activation)



Bomb (after
activation)

6. Game Design

Brick Breaker:

The following screen captures the Brick Breaker arcade game within LearnCade. It involves the user moving a paddle to launch and control the characters trajectory to break down bricks to collect an answer circle. In this case, the correct answer is C. heart, therefore, the user would use the character and the powerups such as the bomb to help clear the bricks that are holding the answer circle, once they are destroyed the circle will drop and then the user must collect it with their paddle to complete the level and earn score. Additionally, we have four +100 powerups that will allow users to earn higher scores at the cost of time; therefore, users must plan and strategize.

If they answer incorrectly by either dropping the right answer circle or collecting the wrong answer circle they will lose a life. Another way they can lose a life is by simply running out of time.

Overall, it effectively combines the traditional Breakout concept with strong educational value and entertainment for the target audience.

In the example shown, the user has chosen Brick Breaker, Science, Human Body, 3 lives, 10 questions/levels, 40 second timer, and a theme of 'red and pink'.

What organ pumps blood around your body?

A. brain B. lungs C. heart D. stomach

Score: 0 Multiplier: 1.0x Lives: 3 Level: 1 / 10 Timer: 36s



7. Revenue Model

LearnCade will have three potential streams to generate income:



1. Subscription plans

LearnCade will be primarily monetised through subscription plans. Once a user signs up and makes an account, they will be assigned a Free basic plan which will offer a limited number of game generations before a cooldown/waiting period. This is essential as it will attract new users and allow them to try the application. If they see it as a valuable tool they can choose to upgrade to a Premium plan. A Premium plan will unlock unlimited game generations, allowing users to create as many games as they would like. This type of monetisation approach will attract the parents of the kids/students as they will want their kids to use LearnCade as an educational tool to achieve better grades.



2. In-App transactions:

Another way LearnCade will generate income is through In-App transactions. LearnCade will aim to offer unique themes, new game templates, character customisation etc, for users to purchase. This approach will mainly attract the kids as they will want the best-looking characters and/or game designs compared to their friends/other students. However, I recognise that kids will not have their own money, therefore the parents are still the main customer base in this approach.



3. Institutional Licenses

Finally, LearnCade can monetise by selling institutional licenses to primary schools and/or other educational partners that educate the intended target audience. The approach would be to sell annual or multi-year plans, which would attract schools as it allows them to easily implement the system into their curriculum, and it would also be cheaper because they wouldn't have to pay for individual subscription plans for each student.